

Course syllabus for

Academic year 2022/2023 >

RRY125 - Modern astrophysics

Modern astrofysik

Course syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [MPPHS](#)**7,5 Credits****Grading:** TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail**Education cycle:** Second-cycle**Main field of study:** Electrical Engineering, Engineering Physics**Department:** 70 - SPACE, EARTH AND ENVIRONMENT

Course round 1

Teaching language: English**Application code:** 85129**Open for exchange students:** Yes**Block schedule:** [C](#)

Maximum participants: 40

Credit distribution

Module				Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0111	Written and oral assignments	1,5 c	Grading: UG		1,5 c					
0211	Examination	6,0 c	Grading: TH		6,0 c					Tue 10/01-2023 am J, Mon 03/04-2023 pm J, Fri 25/08-2023 pm J

In programs

[MPPHS PHYSICS, MSC PROGR, Year 1 \(compulsory elective\)](#)[TKTFY ENGINEERING PHYSICS, Year 3 \(elective\)](#)

Examiner:

[Susanne Aalto](#) [Go to course homepage](#)

Eligibility

General entry requirements for Master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Mathematics 30 c (including multivariable calculus), basic physics(including mechanics, electromagnetism, quantum physics).

Aim

The aim of the course is to introduce the students to the most important concepts of astrophysics and to give an overview of the modern view of the Universe. In particular, the course will focus on how basic properties of cosmic objects are measured, how physics is used in models of cosmic objects, and how astrophysical models are tested with observations. The course will provide a basis for continued studies in astronomy and astrophysics, but will also show students who choose to specialize in other branches of science how physics is used to understand the complex systems that make up our universe.

Learning outcomes (after completion of the course the student should be able to)

- give an overview of the origin, structure and evolution of the Universe and its contents (planets, stars, galaxies),

- explain how basic properties of cosmic objects are measured (e.g. distances, sizes, masses and temperatures),
- explain methods to discover exoplanets,
- perform calculations (based on observational data) of temperatures and ages of solar system bodies, masses and radii of exoplanets and binary stars, magnitudes and luminosities, cosmic distances, masses of galaxies, and the expansion of the Universe,
- explain and use the radiative transfer equation in simple cases
- explain the basic physics and equations of stellar structure,
- review stellar evolution, and relate it to observations (including the Hertzsprung-Russell diagram),
- analyze radioastronomical observations to study the structure and kinematics of the Milky Way
- use kinematics to explain the basic structure of galaxies (spirals, ellipticals),
- review the present cosmological model and the evidence for it (including evidence for dark matter and dark energy),
- derive the Friedmann equation using Newton's mechanics and results from relativity, and solve it in mathematically simple cases

Content

The emphasis is on physical understanding and on principles for how important properties of cosmic objects are measured. Several branches of physics are utilized (mechanics, quantum physics, statistical physics, nuclear physics, electromagnetism). Order of magnitude estimates will be used frequently.

- The Solar system (overview, geology and atmospheres, formation),
- Exoplanets: discovery and properties
- Stars (observational results, stellar structure and evolution, binary stars, compact stars)
- The Milky Way (basic structure, interstellar medium, star clusters, kinematics and structural components)
- Galaxies and galaxy clusters (galaxy classification and observational results, basic galactic kinematics and dynamics, active galactic nuclei, galaxy interactions and galaxy evolution),
- Cosmology (expansion of the universe, abundances of elements, the cosmic microwave background, the big bang, cosmological models)

Organisation

The course includes lectures, problem solving sessions and a compulsory project.

Literature

"Astrophysics for Physicists", Arnab Rai Choudhuri, Cambridge University Press (2010). Also available on-line as an e-book at Chalmers Library.

Examination including compulsory elements

Written examination and compulsory assignment.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.